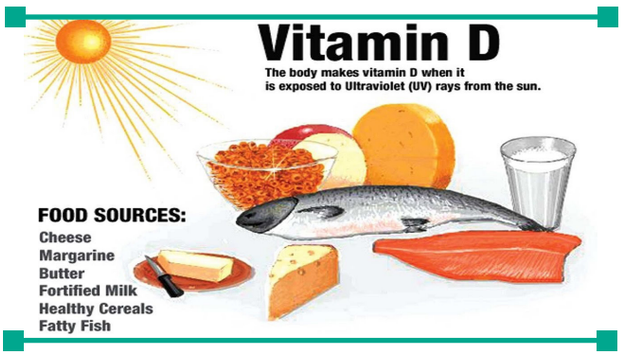
in brain, kidney, liver, skin, reproductive tissues.

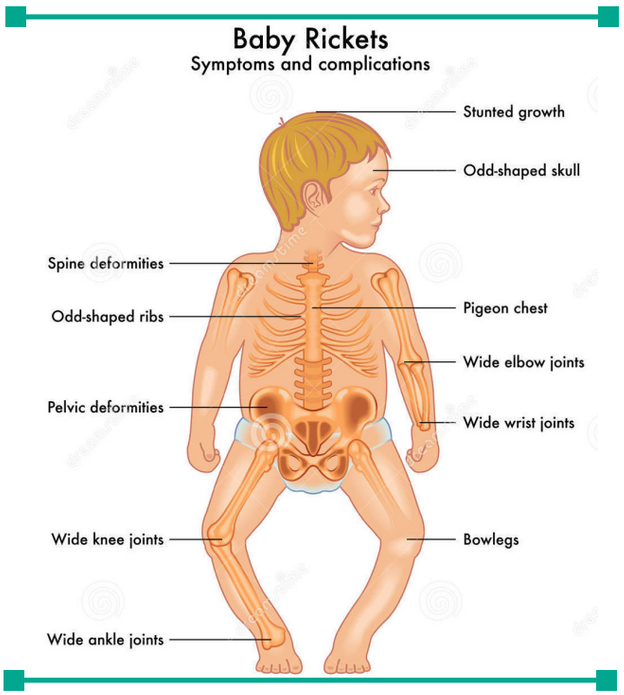
- **Food sources:** Vitamin D is present only in some foods of animal origin. Certain marine fishes and fresh water fishes are known to be good sources of vitamin D. The most important sources are egg yolk, butter, cheese, milk.



- **Deficiency:** Vitamin D deficiency occurs in children who are not adequately exposed to sunlight. It is characterized by inadequate mineralization of the bone. In children the condition is known as rickets and in adults it is called Osteomalacia.

  Rickets: In rickets there is softening of the skull bones and the head is enlarged, elongated and flattened on the vertex. Softening of the ribs, sinking of the chest, beaded junctions of the ribs with cartilages (rickety rosary), pigeon chest, knock knees and bow legs. Deformities of the long bones spine, pelvis, muscles, and feet are observed. Dentition is delayed.

  Osteomalacia: It is the adult counterpart of rickets. It occurs in



women of child bearing age and in those who consume poor cereal diets deficient in vitamin D and calcium. Besides it is found among those who stay indoors all day and seldom go out in the sun.

(c) **Vitamin E:** Vitamin E is the generic name for a group of vitamins, three of which –alpha tocopherol, beta tocopherol and gamma tocopherol display the greatest biologic activity. Of these three, alpha tocopherol is the most significant form of Vitamin E

- **Functions**
  - Vitamin E a major antioxidant which reduces the incidence of heart diseases.
  - It is essential for normal reproduction in man.
  - It acts along with selenium in reducing the body's requirement for each other.
  - It plays a vital role in the immune function of the body.

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